INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Trifluridine/tipiracil (Lonsurf) monotherapy

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
28 days
See the active treatment phase

Treatment stages
1
Explained in the treatment reference

Regimen source
NCCP 00382
Version 4

Two tablet weeks within a 28-day cycle

Treatment stages

01 **Two five-day dosing blocks, then recovery**
28-day cycle

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

		PAGE
01	Understanding your treatment	2
02	Your schedule, tests and appointments	5
03	Symptoms and supportive care	9
04	Your record and contact card	16
05	Sources and further information	19

My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Call your oncology team immediately for a temperature of 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell. Infection can occur without a fever. Call urgently for bleeding, severe diarrhoea, repeated vomiting, inability to drink, very little urine or new breathlessness. Severe breathing difficulty, heavy bleeding or collapse needs 112 / 999.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

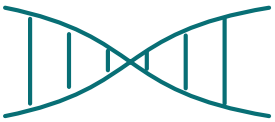
Two ingredients in one chemotherapy tablet

This oral chemotherapy aims to control colorectal cancer that has spread after earlier treatments, or when those treatments are unsuitable. It combines two ingredients in one medicine. Your team discusses the likely benefit, uncertainty, alternatives and your priorities.

ORAL CYTOTOXIC CHEMOTHERAPY

Trifluridine/tipiracil

Lonsurf



Damages cancer-cell deoxyribonucleic acid (the cell's genetic material) (DNA) while slowing breakdown of the active drug

Trifluridine is incorporated into DNA and interferes with cell growth. Tipiracil slows its breakdown so more trifluridine can reach the bloodstream. Healthy cells can also be affected; blood counts and symptoms need monitoring.

Trifluridine enters DNA → Tipiracil slows drug breakdown → Cancer-cell growth is disrupted

What to watch for

Check the exact dosing and rest days. Report fever, sudden illness, vomiting, diarrhoea, bleeding or reduced urine promptly.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

What this treatment is for

Check the treatment aim and exact regimen.

Separate licensed uses

This guide covers Lonsurf on its own for previously treated metastatic colorectal cancer under NCCP 00382a. It does not cover gastric cancer or Lonsurf with bevacizumab. Confirm the exact regimen on your prescription.

How long it continues

Cycles continue while treatment is helping and side effects are acceptable. There is no fixed total number of cycles. Blood results, symptoms, scans and your preferences guide continuation, delays or stopping.

Evidence for colorectal monotherapy

RECOURSE compared Lonsurf plus supportive care with placebo plus supportive care in previously treated metastatic colorectal cancer.

The trial found longer survival with Lonsurf than placebo in this selected population. A group result cannot predict an individual benefit. Gastric treatment and Lonsurf with bevacizumab are different evidence contexts.

CHAPTER 02

Your schedule, tests and appointments

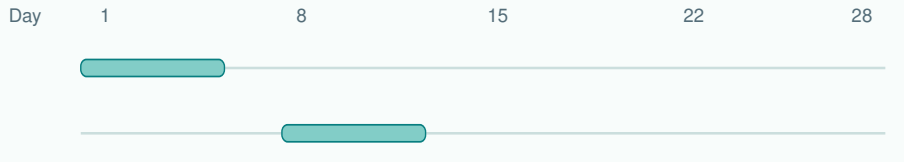
IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

A 28-day tablet cycle

Trifluridine/tipiracil (Lonsurf) monotherapy

28-day view



This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

28-DAY CYCLE

Two five-day dosing blocks, then recovery

D1–5 | D6–7 | D8–12 | D13–28

Take the prescribed morning and evening tablets on Days 1–5, have no Lonsurf on Days 6–7, then take morning and evening tablets on Days 8–12. Days 13–28 are tablet-free. A new 28-day cycle starts only when the team authorises it.

BASELINE → TREATMENT → FOLLOW-UP

Checks through your treatment

Baseline preparation, checks before each cycle and tests arranged because of symptoms are different. No fixed scan interval is imposed here.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00382 v4 · official regimen

Tests in your plan

Carcinoembryonic antigen (CEA) — colorectal plans

If part of your plan

WHEN

For a colorectal cancer plan, if your team uses CEA; your team sets the interval.

A marker trend interpreted alongside symptoms, examination and scans.

Record the baseline, latest result and next date. A marker change alone does not establish progression.

THE WHOLE TREATMENT JOURNEY

Trifluridine/tipiracil (Lonsurf) monotherapy: treatment and checks

Previously treated metastatic colorectal cancer

Take the prescribed morning and evening tablets on Days 1–5, have no Lonsurf on Days 6–7, then take morning and evening tablets on Days 8–12. Days 13–28 are tablet-free. A new 28-day cycle starts only when the team authorises it.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Days 1–5	Days 6–12	Days 13–28	Each next cycle / stopping
Blood count, kidney and liver NCCP 00382	Baseline full blood count (FBC), renal/ liver profile	Extra tests if clinically needed	Closer blood-count monitoring with kidney impairment	Review recovery and new-cycle tests	Repeat before each cycle; follow abnormalities
Urine protein NCCP 00382	Baseline dipstick	Report urinary changes	Additional testing if needed	Review with next-cycle tests	Dipstick before each cycle
Tablets, symptoms and support NCCP 00382	Pharmacy calendar and anti-sickness plan	Check exact AM/PM quantities	No catch-up doses; report symptoms	Rest is still a time to report illness	Reconcile dose history, recovery and support
Treatment benefit NCCP 00382	Individual aim and alternatives	Observe symptoms and function	Contact team if worsening	Scans according to Consultant plan	Continue while helpful and tolerated

What each check is for

1 **Two five-day dosing blocks, then recovery**

28-day cycle

This oral chemotherapy aims to control colorectal cancer that has spread after earlier treatments, or when those treatments are unsuitable. It combines two ingredients in one medicine. Your team discusses the likely benefit, uncertainty, alternatives and your priorities.

Ask / act: Confirm the current phase, prescription, tests and support medicines.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Prescription and supply

Before collecting another course, confirm the dose, tablet strengths, calendar, blood-test date and authorisation. Bring the current packs and medicines list.

STEP 2

Meal routine

Swallow the prescribed tablets with water within one hour after finishing breakfast and the evening meal on dosing days. Doses are based on body size and previous side effects; two tablet strengths may be combined. Some prescriptions have different morning and evening tablet quantities. Use the pharmacy calendar and do not calculate doses yourself.

STEP 3

Missed or vomited doses

Do not make up a missed, held or vomited dose and do not double the next dose. Tell your team and follow the next scheduled dose on the agreed calendar. Repeated vomiting needs advice because of dehydration and missed medicines.

STEP 4

Appointments

Ask who books extra tests, who reviews results and whom to call if the plan is unclear. There is no infusion visit or take-home pump in monotherapy.

STEP 5

Swallowing and handling

Swallow tablets whole with water and wash your hands after handling them. If you cannot swallow tablets whole, ask the pharmacist before changing them. Keep them out of children’s reach and ask the pharmacist about safe disposal.

STEP 6

Accidental extra tablets

If you take more Lonsurf than prescribed, contact a doctor or go to hospital straight away. Take the packs and current calendar with you.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Day 1 → Day 28: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
Lonsurf: first block	 **Days 1–5** Morning and evening tablets after meals
Lonsurf: second block	 **Days 8–12** Morning and evening tablets after meals
First rest	 **Days 6–7** No Lonsurf
Longer rest	 **Days 13–28** No Lonsurf; recovery and review
Report symptoms	 **Days 1–28** Urgent symptoms can occur throughout

Before the first tablet: Complete consent, baseline tests and pharmacy teaching; confirm morning/evening tablet quantities.

Duration: Cycles continue while treatment is helping and side effects are acceptable. There is no fixed total number of cycles. Blood results, symptoms, scans and your preferences guide continuation, delays or stopping.

No catch-up doses: Missed, held or vomited doses are not replaced on dosing or rest days.

> **Why these bands stay broad:** This is a reference schedule. It does not identify safe days or predict when side effects will occur. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment.
> NCCP 00382 v4 · official regimen ↗

> **There are no guaranteed "safe days".** Use your own oncology unit's 24-hour contact plan. Urgent symptoms override the diary; do not wait for the next appointment.

Between-treatment notes

Day 1–5: First tablet block

Take both prescribed daily doses after meals. Follow the exact label and calendar.

Day 6–7: Two rest days

No Lonsurf tablets. Do not use rest days to make up missed doses.

Day 8–12: Second tablet block

Take prescribed morning and evening doses after meals. Call if symptoms or dosing are uncertain.

Day 13–28: Recovery and next-cycle assessment

No Lonsurf tablets. Tests and review guide the next cycle. Do not restart before authorisation.

Support around the medicines

1 A written tablet calendar

2 Prevent and treat sickness

This oral medicine has moderate-to-high sickness risk. Agree a preventive and rescue anti-sickness plan with the team. If you cannot keep fluids or medicines down, call; do not add or repeat doses yourself.

3 Food, fluids and bowel symptoms

Take tablets after meals as instructed. Drink as your team advises and report diarrhoea, repeated vomiting, poor intake or reduced urine early. A prescribed bowel treatment plan does not replace urgent assessment.

4 Blood tests and infection

Blood count, kidney/liver tests and urine-protein checks occur before treatment and each cycle; extra tests may be needed, especially with kidney impairment. Report fever or illness even on rest days.

5 Pregnancy prevention

Discuss fertility before treatment. Women need highly effective contraception during treatment and for six months after; add a barrier method to hormonal contraception. Men with a partner who could become pregnant need effective contraception during treatment and for six months afterwards. Do not breastfeed while taking this medicine.

Your tablet and rest-day plan

A prescription calendar prevents dosing errors.

Take tablets after meals

Tests and dose review

The team authorises each new cycle.

Baseline and each cycle

Blood count, kidney and liver tests and urine-protein screening are required. Additional blood counts may be needed during treatment, especially with kidney impairment.

Why a new cycle may wait

The blood-count requirements for starting another cycle are different from the thresholds for stopping tablets during a cycle. Symptoms, prior severe low counts, time to recover and the current prescription all matter.

Dose changes

Your team may delay treatment or use a lower dose after toxicity. Do not restart from a diary or increase the dose after a reduction. Tablet quantities are checked by pharmacy.

Support at home

Keep contact details and medicines together.



A written tablet calendar



Other medicines

Tell the pharmacist about all prescribed, non-prescription and herbal medicines, including antivirals. Keep tablets safely in their supplied packaging and follow disposal advice.

Safe handling and individual needs

Share any difficulties before the next prescription.



Accidental extra tablets

Lactose and rare inherited conditions

Lonsurf contains lactose. Tell the prescriber about rare inherited galactose intolerance, total lactase deficiency or glucose-galactose malabsorption before treatment; these conditions affect suitability. If you have been told you cannot tolerate some sugars, ask the pharmacist.



Additional blood tests

Extra blood-count tests may be needed during a cycle, particularly with kidney impairment or symptoms. The NCCP regimen requires tests before every cycle or as indicated; it does not specify a mandatory Day 15 test for everyone. Follow your team's individual appointment schedule.

After stopping treatment

Ask when unresolved symptoms or abnormal blood tests will be checked again, who remains your contact and what the next treatment or supportive-care plan is. Continue the prescribed contraception precautions after the last dose.

Questions worth asking

Does having more side effects mean treatment is working?

No. Side effects do not reliably show whether the cancer is responding. Report them so they can be treated. Your team assesses benefit using the tests and review plan appropriate to your cancer.

Why might a dose be delayed or reduced?

Your team may allow recovery, investigate a symptom or reduce the chance of lasting harm. Ask which drug changes, why, and when review is planned. A delay does not itself mean treatment has failed.

What should I say when calling?

Give the regimen name, last treatment date, symptoms and when they started. Explain whether you can eat, drink, pass urine and manage usual activities. Give your temperature if available, but do not delay calling to collect details.

What belongs in the consent conversation?

The individual aim and likely benefit, uncertainty, reasonable alternatives including not proceeding, common and serious risks, lasting effects, fertility and practical concerns. You can ask for an interpreter, more time or another explanation and can reconsider your decision.

How many cycles should I use this passport for?

The passport provides reusable recording pages, not a prescribed course length. Your oncology team sets the duration and can provide extra pages.

Can I use this page to decide if blood results are safe?

No. Treatment decisions require the clinical protocol, symptoms, previous toxicity, organ function and the prescriber's review. The patient page explains the checks and does not authorise treatment.

Are tablet-free days safe days?

No. The rest period allows recovery, but low blood counts and other problems can occur then too. Use the same urgent contact plan throughout the cycle.

Can my morning and evening tablet numbers differ?

Yes. The pharmacy may prescribe different morning and evening quantities to match the correct total daily dose. Do not assume the same number or substitute tablet strengths; follow the written label and calendar.

Is the bevacizumab combination evidence the same?

No. A study of Lonsurf plus bevacizumab tests an additional medicine and its extra risks. Its results should not be presented as the expected benefit of Lonsurf on its own.

> **Try explaining the plan back:** Can you show the two five-day dosing blocks and the rest days, explain how you take tablets with meals, what you do after a missed or vomited dose, and how you get urgent help?

Treatment scope · NCCP 00382 This guide covers Lonsurf on its own for previously treated metastatic colorectal cancer under NCCP 00382a. It does not cover gastric cancer or Lonsurf with bevacizumab. Confirm the exact regimen on your prescription.

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

Cyclic oral monotherapy: moderate-to-high emetic risk

Active medicines: trifluridine_tipiracil

Emetogenic risk: moderate_to_high

Medicine	Dose / route	When
One oral 5-HT3 antagonist (choose one)	NCCP oral reference options: granisetron 1–2 mg total daily OR ondansetron 8–16 mg total daily	Start before oral treatment and continue daily according to the prescribed course; team specifies dosing and rest-day coverage Alternatives, not concurrent prescriptions. Assess QT risk, constipation, interacting medicines, organ function and local formulary. No routine added steroid is inferred from parenteral guidance.

Additional phase-specific support

- Agree oral preventive antiemetic and rescue instructions under NCCP oral guidance/local policy; this is not the NCIS parenteral moderate/high protocol.
- No routine premedication is specified by NCCP 00382.
- Assess need for granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) or other infection support individually; no universal prophylaxis is prescribed here.
- Use the exact dosing calendar and no catch-up doses.
- Reconcile gastrointestinal (GI) symptoms, hydration and additional blood counts for renal impairment.
- Confirm the body-surface-area prescription, actual morning and evening tablet counts and Days 1–5 and 8–12 calendar. Take within one hour after breakfast and the evening meal; do not replace missed or vomited doses.
- Follow current NCCP cycle-start blood and urine checks and the additional EMA blood-count requirement before starting and as needed, at minimum before each cycle; increase monitoring with renal impairment or toxicity.
- Prior reductions must be retained: no dose re-escalation. The severe-renal dosing pathway differs from the standard minimum dose and needs separate review.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

What to watch for

You do not need to know which medicine caused a symptom before calling.

1 Energy / wellbeing

Trifluridine/tipiracil

Fatigue, dizziness or sudden illness: report changes.

3 Blood counts

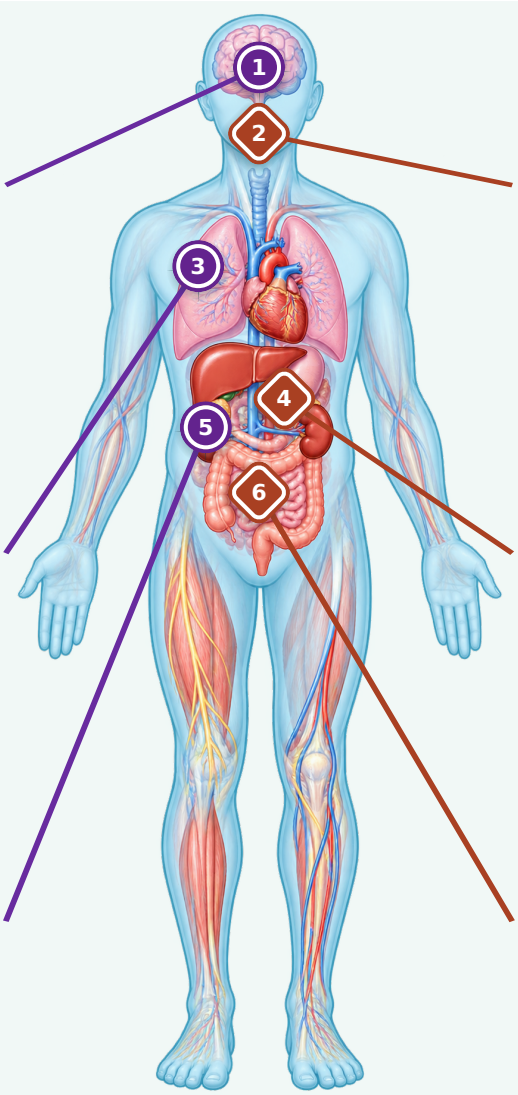
Trifluridine/tipiracil

Infection, anaemia or bleeding may occur; blood tests and urgent symptom reporting matter.

5 Kidneys / fluids

Trifluridine/tipiracil

Reduced urine or dehydration needs review; kidney impairment can increase drug toxicity.



2 Mouth

Trifluridine/tipiracil

Soreness or ulcers can affect eating and drinking.

4 Sickness / appetite

Trifluridine/tipiracil

Nausea, vomiting or poor intake can cause dehydration.

6 Bowel

Trifluridine/tipiracil

Diarrhoea, cramps or severe tummy pain: call early.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Call your oncology team immediately for a temperature of 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell. Infection can occur without a fever. Call urgently for bleeding, severe diarrhoea, repeated vomiting, inability to drink, very little urine or new breathlessness. Severe breathing difficulty, heavy bleeding or collapse needs 112 / 999.

When to get help

Call 112 / 999 for severe breathing difficulty, severe chest pain, collapse or a seizure.

Call oncology immediately: temperature 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell, even with a normal temperature. Follow your hospital's temperature instructions. If you cannot reach the team, seek urgent medical assessment. Do not wait for the next appointment.

Oncology daytime: 087 12345678 · **Out of hours:** 087 87654321

EXAMPLE ONLY — team must replace both numbers before patient use.

CHAPTER 04

Your record and contact card

IN THIS CHAPTER

Space for your own cycle or review record, questions and notes, followed by your treatment card.

Record each cycle separately. Each medicine has its own symptom table. Mark the worst symptoms since your last review, when they occurred and what they stopped you doing. Overlapping symptoms can be recorded under more than one medicine; you do not have to identify the cause.

These are patient-language severity prompts, not a validated questionnaire or a formal Common Terminology Criteria for Adverse Events (CTCAE) grading tool. Your team assigns the clinical grade. Serious symptoms override the columns: call without waiting to decide a severity level.

Use a new record sheet for each cycle or review. Copy or print another sheet when needed; the number of sheets does not prescribe the length of treatment.

Cycle number: _____ Start date: _____ Next review: _____

Recording pages are reusable and do not prescribe a number of cycles or authorise a restart.

Trifluridine/tipiracil · Oral cytotoxic chemotherapy

Symptom	None	Mild	Moderate / call	Severe / urgent	Emergency signs
Nausea	□ None	□ Queasy but eating and drinking	□ Intake reduced: contact the team	□ Unable to maintain intake: urgent review	□ Collapse / severe dehydration: 112/999
Diarrhoea	□ Usual pattern	□ Up to 3 extra stools above normal	□ 4–6 extra stools/ day or at night: call	□ 7+ extra stools/ day or dehydration: urgent review	□ Collapse / severe dehydration: 112/999
Fatigue	□ Usual energy	□ Tiredness improves with rest	□ Rest does not help; usual tasks affected: call	□ Self-care difficult: urgent review	□ Collapse: 112/999
Mouth soreness / ulcers	□ None	□ Mild soreness; eating normally	□ Painful; soft food needed: call	□ Cannot eat / drink enough: urgent review	□ Major breathing difficulty: 112/999
Bleeding / bruising	□ None	□ New small bruise / brief bleed: report	□ Repeated bleeding: call today	□ Bleeding that will not stop: urgent help	□ Heavy bleeding / collapse: 112/999

Dates / doses missed or changed / symptoms affecting daily life / questions:

Red flags override the table. Fever, shivers, suddenly unwell, chest symptoms or inability to drink: seek prompt advice. Severe breathing difficulty, collapse or severe chest pain: 112 / 999.

Keep your treatment details to hand

Previously treated metastatic colorectal cancer · NCCP 00382 v4 · Document revised 5 November 2026

Print this page **one-sided on A4 at Actual size / 100%**. Cut only the dashed outer border around both panels. **Do not cut the centre line.** Fold with FRONT and BACK facing out, then glue or tape the blank backs together. Finished size: **85.6 × 54 mm**. Keep the stamp side writable.

FRONT / CONTACTS

My Treatment Card

SACTCheck

Name: _____

Lonsurf (trifluridine/tipiracil)

28 days · tablets twice daily Days 1–5 and 8–12 · remaining days tablet-free.

Phase: _____ Cycle plan: team to confirm

Daytime (EXAMPLE) 087 12345678

Out of hours (EXAMPLE) 087 87654321

Unwell or fever? Call oncology now.
Severe breathing difficulty, chest pain or collapse: **112 / 999.**

EXAMPLE NUMBERS — replace before patient use.

BACK / TREATMENT RECORD

My cycle record

Rolling record — follow your agreed treatment plan.

Cycle	Date / initials

My treatment information

Latest treatment date: _____

Plan changed? Ask the team to update this card.

CUT outer border

FOLD centre — do not cut

Keep the record accurate

The team dates and initials each recorded treatment unit when complete. A cycle is complete after its final planned dose; a weekly-treatment stamp records that weekly dose. Delays, omitted doses or a changed plan need a note or a replacement card. A stamp does not approve the next treatment. Keep any medicine-specific alert card too.

Put your treatment page on your phone

Scan your QR and check the treatment name. Save that page to your home screen.

iPhone / Safari

Tap Share → Add to Home Screen. Select Open as Web App if offered, then Add.

Android / Chrome

Tap the three-dot menu → Add to Home screen or Install and create shortcut. Choose the shortcut option and confirm.

Menu wording varies. The QR opens information, not a personal record. Paper stamps do not update the website. Check that the saved icon opens the correct treatment.

Your local contacts and resources

Oncology daytime: 087 12345678 · **Out of hours:** 087 87654321

EXAMPLE ONLY — team must replace both numbers before patient use.

Hospital / team: _____
Backup if the team cannot be reached: _____

Additional local services and resources will be added when confirmed. Your team must check and complete this page before patient use.

SACTCheck v0.84.34 · NCCP 00382 v4 · Wallet-card sheet
Feasibility build. Clinical, pharmacy and patient/PPI review pending. Your team confirms your personal plan.

CHAPTER 05

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **National Cancer Control Programme (NCCP) 00382 v4 · official regimen**
   https://healthservice.hse.ie/documents/6348/trifluridine_and_tipiracil_Lonsurf_therapy_382.pdf
2. **EMA Lonsurf · current product information and patient leaflet · checked 5 November 2026**
   https://www.ema.europa.eu/en/documents/product-information/lonsurf-epar-product-information_en.pdf
3. **NCCP antiemetic classification V6: oral medicines**
   https://healthservice.hse.ie/documents/7152/Classification_Document_for_Systemic_Anti-_Cancer_Therapy_systemic anticancer therapy (SACT)_Induced_Nausea__pWgDCrm.pdf
4. **NCCP NCIS Medical Oncology antiemetics V8 (parenteral context)**
   https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
5. **RECOURSE final survival analysis, 2018 publication**
   https://pubmed.ncbi.nlm.nih.gov/29274618/
6. **HSE chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
7. **NCI: colon cancer treatment — carcinoembryonic antigen (CEA) and follow-up**
   https://www.cancer.gov/types/colorectal/hp/colon-treatment-pdq